

求积分

$$\int_0^1 x \bmod x^2 \mathrm{d}x$$

由恒等式

$$x \bmod y = x - y \left\lfloor \frac{x}{y} \right\rfloor$$

那么

$$\begin{aligned} & \int_0^1 x \bmod x^2 \mathrm{d}x \\ &= \int_0^1 x - x^2 \left\lfloor \frac{1}{x} \right\rfloor \mathrm{d}x \\ &= - \sum_{n=1}^{\infty} \int_{\frac{1}{n}}^{\frac{1}{n+1}} x - x^2 \left\lfloor \frac{1}{x} \right\rfloor \mathrm{d}x \\ &= - \sum_{n=1}^{\infty} \int_{\frac{1}{n}}^{\frac{1}{n+1}} x - nx^2 \mathrm{d}x \\ &= \sum_{n=1}^{\infty} \frac{1}{2} \left[\frac{1}{n^2} - \frac{1}{(n+1)^2} \right] - \frac{n}{3} \left[\frac{1}{n^3} - \frac{1}{(n+1)^3} \right] \\ &= \frac{1}{2} - \frac{1}{3} \sum_{n=1}^{\infty} n \left(\frac{1}{n^3} - \frac{1}{(n+1)^3} \right) \\ &= \frac{1}{2} - \frac{1}{3} \sum_{n=1}^{\infty} \frac{1}{n^3} \\ &= \frac{1}{2} - \frac{1}{3} \zeta(3) \end{aligned}$$